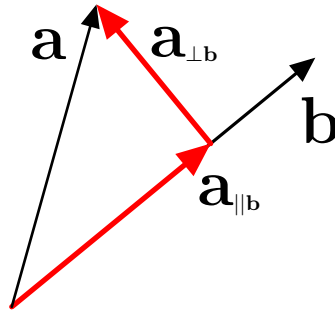


Previously we have used geometric algebra to mainly clarify concepts from analysis. Here we turn our attention to making unoriented measurements of the separation between lines. An application of counting equivalence classes of non-null similar triangles over finite fields is given.

Given two vectors $\mathbf{a}$ and $\mathbf{b}$ in $GA(\mathbb{V}, Q(\mathbf{x}))$ we frequently wish to be able to decompose $\mathbf{a}$ into parts that are parallel and perpendicular to $\mathbf{b}$, using the notation $\mathbf{a} = \mathbf{a}_{\parallel \mathbf{b}} + \mathbf{a}_{\perp \mathbf{b}}$. We call $\mathbf{a}_{\parallel \mathbf{b}}$ the **projection** of $\mathbf{a}$ onto $\mathbf{b}$, and $\mathbf{a}_{\perp \mathbf{b}}$ the **rejection** of $\mathbf{a}$ from $\mathbf{b}$.



In a usual course on vector based geometry, we would start off finding a formula for the projection by subtracting it from both sides of our equality $\mathbf{a} - \mathbf{a}_{\parallel \mathbf{b}} = \mathbf{a}_{\perp \mathbf{b}}$. Then using the facts that our projection is parallel to $\mathbf{b}$ to write $\mathbf{a}_{\parallel \mathbf{b}} = \lambda \mathbf{b}$ for some scalar λ , and that our rejection is perpendicular to $\mathbf{b}$, we can say that

$$(\mathbf{a} - \lambda \mathbf{b}) \cdot \mathbf{b} = 0.$$

Solving for λ we get $\lambda = (\mathbf{a} \cdot \mathbf{b})/\mathbf{b}^2$ and so

$$\mathbf{a}_{\parallel \mathbf{b}} = \frac{(\mathbf{a} \cdot \mathbf{b})}{\mathbf{b}^2} \mathbf{b}.$$

In the geometric algebra of our plane we can reinterpret this formula as $\mathbf{a}_{\parallel \mathbf{b}} = (\mathbf{a} \cdot \mathbf{b})\mathbf{b}^{-1}$, which suggests there should be something deeper involving vector division going on here. It is almost unfair how effective the geometric product is for this problem. The first few steps are just using the existence of vector multiplicative inverses (we cannot define these concepts for $\mathbf{b}$ a null vector $\mathbf{b}^2 = 0$), associativity, and the fundamental identity

$$\mathbf{a} = \mathbf{a}(\mathbf{b}\mathbf{b}^{-1}) = (\mathbf{a}\mathbf{b})\mathbf{b}^{-1} = (\mathbf{a} \cdot \mathbf{b} + \mathbf{a} \wedge \mathbf{b})\mathbf{b}^{-1}.$$

If we then use the distributive law we get

$$\mathbf{a} = (\mathbf{a} \cdot \mathbf{b})\mathbf{b}^{-1} + (\mathbf{a} \wedge \mathbf{b})\mathbf{b}^{-1}$$

and all that is left to do is interpret.

It may not be immediately clear based on what we have done up to this point that $(\mathbf{a} \wedge \mathbf{b})\mathbf{b}^{-1}$ is actually a vector, though there are various algebraic proofs that it is true. The easiest two is probably just by rearrangement by subtraction $(\mathbf{a} \wedge \mathbf{b})\mathbf{b}^{-1} = \mathbf{a} - \mathbf{a}(\mathbf{a} \cdot \mathbf{b})\mathbf{b}^{-1}$ and using the fact that $\mathbb{V}$ is closed under vector addition, or by writing this expression in your favorite coordinate systems and simplifying. Once you have convinced yourself that we are dealing with vectors only, you may also notice that we have accidentally reproduced the formula for the projection as one of the two terms on our expression for $\mathbf{a}$ decomposed against $\mathbf{b}$. That must mean that $\mathbf{a}_{\perp \mathbf{b}} = (\mathbf{a} \wedge \mathbf{b})\mathbf{b}^{-1}$.

Exercise 1 Show that $\mathbf{a}' = \mathbf{b}\mathbf{a}\mathbf{b}^{-1}$ is the reflection of $\mathbf{a}$ in $\mathbf{b}$. Demonstrate that this transformation fixes quadrances $(\mathbf{a}_1 - \mathbf{a}_2)^2 = (\mathbf{a}'_1 - \mathbf{a}'_2)^2$ and flips the sign of wedge products $\mathbf{a}_1 \wedge \mathbf{a}_2 = -\mathbf{a}'_1 \wedge \mathbf{a}'_2$.

In 2005 [Wildberger introduced rational trigonometry](#). He wanted to create a framework for affine geometry which was agnostic of the underlying field and quadratic form chosen to study shapes with. To accomplish this task he stressed the importance of quadrance to measure the separation between points, and he introduced several oriented and unoriented quantities to measure the separation between lines. Trying to understand the source of these measurements and the convenient identities which govern their use in a more unified manner was my first impetuous for investigating geometric algebra more deeply.

Given two vectors $\mathbf{a}, \mathbf{b}$ define their **cross** to be $c(\mathbf{a}, \mathbf{b}) = \mathbf{a}_{\parallel \mathbf{b}}^2 / \mathbf{a}^2$ and similarly their **spread** to be $s(\mathbf{a}, \mathbf{b}) = \mathbf{a}_{\perp \mathbf{b}}^2 / \mathbf{a}^2$. At first glance it may not seem like these are helpful quantities to examine. Any reasonable unoriented measurements of the separation of lines should be well defined $c(\alpha \mathbf{a}, \beta \mathbf{b}) = c(\mathbf{a}, \mathbf{b})$, $s(\alpha \mathbf{a}, \beta \mathbf{b}) = s(\mathbf{a}, \mathbf{b})$ and symmetric $c(\mathbf{a}, \mathbf{b}) = c(\mathbf{b}, \mathbf{a})$, $s(\mathbf{a}, \mathbf{b}) = s(\mathbf{b}, \mathbf{a})$, but from our first definitions it may not be immediately clear that these properties hold.

Lemma If $\mathbf{c}$ is in the span of $\mathbf{a}$ and $\mathbf{b}$, then $(\mathbf{a} \wedge \mathbf{b})\mathbf{c} = -\mathbf{c}(\mathbf{a} \wedge \mathbf{b})$.

Proof Since $\mathbf{c}$ is in the span of $\mathbf{a}$ and $\mathbf{b}$ we can write $\mathbf{c} = \alpha \mathbf{a} + \beta \mathbf{b}$. Then we have

$$(\mathbf{a} \wedge \mathbf{b})\mathbf{c} = \frac{1}{2}(\mathbf{a}\mathbf{b} - \mathbf{b}\mathbf{a})(\alpha \mathbf{a} + \beta \mathbf{b}) = \frac{1}{2}[\alpha(\mathbf{a}\mathbf{b}\mathbf{a} - \mathbf{b}\mathbf{a}^2) + \beta(\mathbf{a}\mathbf{b}^2 - \mathbf{b}\mathbf{a}\mathbf{b})].$$

Using the fact that scalars commute with everything we can move the squared terms to the left of the vectors

$$(\mathbf{a} \wedge \mathbf{b})\mathbf{c} = \frac{1}{2}[\alpha(\mathbf{a}\mathbf{b}\mathbf{a} - \mathbf{a}^2\mathbf{b}) + \beta(\mathbf{b}^2\mathbf{a} - \mathbf{b}\mathbf{a}\mathbf{b})] = \frac{1}{2}[\alpha\mathbf{a}(\mathbf{b}\mathbf{a} - \mathbf{a}\mathbf{b}) + \beta\mathbf{b}(\mathbf{b}\mathbf{a} - \mathbf{a}\mathbf{b})] = (\alpha\mathbf{a} + \beta\mathbf{b})(\mathbf{b}\mathbf{a} - \mathbf{a}\mathbf{b})$$

which is $-\mathbf{c}(\mathbf{a} \wedge \mathbf{b})$ as required.

Theorem 1 The cross and spread of $\mathbf{a}$ and $\mathbf{b}$ have the following formulas

$$c(\mathbf{a}, \mathbf{b}) = \frac{(\mathbf{a} \cdot \mathbf{b})^2}{\mathbf{a}^2 \mathbf{b}^2}, \quad s(\mathbf{a}, \mathbf{b}) = \frac{-(\mathbf{a} \wedge \mathbf{b})^2}{\mathbf{a}^2 \mathbf{b}^2},$$

which proves they are both well defined and symmetric.

Proof The proof is just plugging in the formulas we found in the beginning into our definitions. For cross it is straightforward

$$\mathbf{a}_{\parallel \mathbf{b}}^2 / \mathbf{a}^2 = \frac{[(\mathbf{a} \cdot \mathbf{b})\mathbf{b}^{-1}]^2}{\mathbf{a}^2} = \frac{(\mathbf{a} \cdot \mathbf{b})^2 \mathbf{b}^2}{\mathbf{a}^2 \mathbf{b}^4} = \frac{(\mathbf{a} \cdot \mathbf{b})^2}{\mathbf{a}^2 \mathbf{b}^2}.$$

For spread we will need our lemma

$$\mathbf{a}_{\perp \mathbf{b}}^2 / \mathbf{a}^2 = \frac{[(\mathbf{a} \wedge \mathbf{b})\mathbf{b}^{-1}]^2}{\mathbf{a}^2} = \frac{(\mathbf{a} \wedge \mathbf{b})\mathbf{b}(\mathbf{a} \wedge \mathbf{b})\mathbf{b}}{\mathbf{a}^2 \mathbf{b}^4} = \frac{-(\mathbf{a} \wedge \mathbf{b})^2}{\mathbf{a}^2 \mathbf{b}^2}.$$

Demonstrating that these formulas imply that our measurements are well defined and symmetric is left to the reader.

Exercise 2 What are the formulas for cross and spread in terms of the coordinates for blue, red, and green geometry? Prove that $c(\mathbf{a}, \mathbf{b}) + s(\mathbf{a}, \mathbf{b}) = 1$. There are at least two possible proofs using geometric algebra. Conclude that whenever we have vectors $\mathbf{a}, \mathbf{b}$ with coefficients in $\mathbb{F}$, then $\cos^2(\theta_B(\mathbf{a}, \mathbf{b})), \sin^2(\theta_B(\mathbf{a}, \mathbf{b})), \cosh^2(\theta_R(\mathbf{a}, \mathbf{b})), \sinh^2(\theta_R(\mathbf{a}, \mathbf{b})) \in \mathbb{F}$ if these quantities are defined. Notice that over a linearly ordered field cross and spread can be defined for vectors that square to opposite signs. Over a linearly ordered field, what is the range of possible values for cross and spread in blue, red, and green geometry?

These measurements of separation between lines seemingly have a defect in comparison to angles: they are not additive. Given a triangle ABC with boundary $\mathbf{a} = C - B$, $\mathbf{b} = A - C$, $\mathbf{c} = B - A$ one of the first thing that school children learn about it is that $\theta_B(\mathbf{a}, \mathbf{b}) + \theta_B(\mathbf{b}, \mathbf{c}) + \theta_B(\mathbf{c}, \mathbf{a}) = \pi$ the sum of the angles in a triangle is always constant. Angles lead to systems of linear equations, as well as necessary and sufficient conditions for a quadrilateral to be cyclic. This situation with angles is not as favorable as it may seem. It may not be a problem for everyone but mathematicians care when an operation requires passing to a field extension because it increases the difficulty of arithmetic or introduces approximations. It is also the case that proofs with angles often require casework on different configurations to be complete, unless directed angles are introduced. It is also not clear to me how to compute an angle over a field with positive characteristic.

The healthy perspective is that every tool that can solve a problem is valuable, and it is worthwhile to make your knowledge redundant by appreciating multiple perspectives. With that in mind, is there a relationship between the spreads of a triangle? Can we use this relationship to solve problems that we couldn't before? The answer is yes to both of these questions, but we need two results which are proper theorems in their own right. The first is what Wildberger refers to as the *cross law* but should be familiar to most as the square of the law of cosines.

Theorem 2 Given a non-null (no sides squaring to zero) triangle ABC with boundary $\mathbf{a} = C - B$, $\mathbf{b} = A - C$, $\mathbf{c} = B - A$ then $4\mathbf{a}^2 \mathbf{b}^2 c(\mathbf{a}, \mathbf{b}) = (\mathbf{a}^2 + \mathbf{b}^2 - \mathbf{c}^2)^2$. In particular, if $\mathbf{a} \perp \mathbf{b}$ then $\mathbf{a}^2 + \mathbf{b}^2 = \mathbf{c}^2$.

Proof From the boundary condition $\mathbf{a} + \mathbf{b} + \mathbf{c} = 0$ we can isolate $\mathbf{c}$ and square everything to get $\mathbf{a}^2 + 2\mathbf{a} \cdot \mathbf{b} + \mathbf{b}^2 = \mathbf{c}^2$. Subtracting our squared vectors to one side we get $2\mathbf{a} \cdot \mathbf{b} = \mathbf{c}^2 - \mathbf{a}^2 - \mathbf{b}^2$ and squaring again leaves us with $4(\mathbf{a} \cdot \mathbf{b})^2 = (\mathbf{c}^2 - \mathbf{a}^2 - \mathbf{b}^2)^2$. Recognizing $(\mathbf{a} \cdot \mathbf{b})^2 = \mathbf{a}^2 \mathbf{b}^2 c(\mathbf{a}, \mathbf{b})$ completes the proof.

Exercise 3 Show that if $\mathbf{a} \parallel \mathbf{b}$ and $\mathbf{c} = \mathbf{a} \pm \mathbf{b}$ then $(\mathbf{a}^2 + \mathbf{b}^2 + \mathbf{c}^2)^2 = 2(\mathbf{a}^4 + \mathbf{b}^4 + \mathbf{c}^4)$. Wildberger calls this the *triple quad formula*, and it shows us that dot is also capable of detecting colinearity.

Exercise 4 This exercise is a fully optional list of identities that are useful to discussions in the future and the next exercise. Define the *Archimedes function* to be $\mathcal{A}(\mathbf{a}, \mathbf{b}, \mathbf{c}) = (\mathbf{a}^2 + \mathbf{b}^2 + \mathbf{c}^2)^2 - 2(\mathbf{a}^4 + \mathbf{b}^4 + \mathbf{c}^4)$. Show that

$$\mathcal{A}(\mathbf{a}, \mathbf{b}, \mathbf{c}) = 4 \det \begin{bmatrix} \mathbf{a}^2 & \mathbf{a} \cdot \mathbf{b} \\ \mathbf{a} \cdot \mathbf{b} & \mathbf{b}^2 \end{bmatrix} = 4 \det \begin{bmatrix} \mathbf{b}^2 & \mathbf{b} \cdot \mathbf{c} \\ \mathbf{b} \cdot \mathbf{c} & \mathbf{c}^2 \end{bmatrix} = 4 \det \begin{bmatrix} \mathbf{c}^2 & \mathbf{c} \cdot \mathbf{a} \\ \mathbf{c} \cdot \mathbf{a} & \mathbf{a}^2 \end{bmatrix},$$

(what does the Cauchy Binet theorem say about these determinants?)

$$\mathcal{A}(\mathbf{a}, \mathbf{b}, \mathbf{c}) = (\mathbf{b}^2 + \mathbf{c}^2 - \mathbf{a}^2)\mathbf{a}^2 + (\mathbf{c}^2 + \mathbf{a}^2 - \mathbf{b}^2)\mathbf{b}^2 + (\mathbf{a}^2 + \mathbf{b}^2 - \mathbf{c}^2)\mathbf{c}^2,$$

$$\mathcal{A}(\mathbf{a}, \mathbf{b}, \mathbf{c}) = (\mathbf{b}^2 + \mathbf{c}^2 - \mathbf{a}^2)(\mathbf{c}^2 + \mathbf{a}^2 - \mathbf{b}^2) + (\mathbf{c}^2 + \mathbf{a}^2 - \mathbf{b}^2)(\mathbf{a}^2 + \mathbf{b}^2 - \mathbf{c}^2) + (\mathbf{a}^2 + \mathbf{b}^2 - \mathbf{c}^2)(\mathbf{b}^2 + \mathbf{c}^2 - \mathbf{a}^2).$$

The second theorem we will need is called the *spread law*.

Theorem 3 In triangle ABC with boundary $\mathbf{a} = C - B$, $\mathbf{b} = A - C$, $\mathbf{c} = B - A$ then and spreads $s_A = s(\mathbf{b}, \mathbf{c})$, $s_B = s(\mathbf{c}, \mathbf{a})$, and $s_C = s(\mathbf{a}, \mathbf{b})$ then

$$\frac{s_A}{\mathbf{a}^2} = \frac{s_B}{\mathbf{b}^2} = \frac{s_C}{\mathbf{c}^2}.$$

Proof By wedging our boundary condition $\mathbf{a} + \mathbf{b} + \mathbf{c} = 0$ with $\mathbf{b}$ and $\mathbf{c}$ we can see that $\mathbf{a} \wedge \mathbf{b} = \mathbf{b} \wedge \mathbf{c} = \mathbf{c} \wedge \mathbf{a}$. By squaring everything and dividing by $-\mathbf{a}^2 \mathbf{b}^2 \mathbf{c}^2$ we get the result.

Exercise 5 The following exercise is highly nontrivial but lies at the heart of the differences between spread and angles. Given a non-null non-colinear triangle ABC show there is a unique point O in the plane determined by ABC , called the **circumcircle** of ABC , such that $(A - O)^2 = (B - O)^2 = (C - O)^2 = K$. Find a formula for O in the barycentric coordinates of ABC expressed using only the quadrances of the sides and the Archimedes function. Prove that the square of a trivector is

$$(\mathbf{a} \wedge \mathbf{b} \wedge \mathbf{c})^2 = \det \begin{bmatrix} \mathbf{a}^2 & \mathbf{a} \cdot \mathbf{b} & \mathbf{a} \cdot \mathbf{c} \\ \mathbf{b} \cdot \mathbf{a} & \mathbf{b}^2 & \mathbf{b} \cdot \mathbf{c} \\ \mathbf{c} \cdot \mathbf{a} & \mathbf{c} \cdot \mathbf{b} & \mathbf{c}^2 \end{bmatrix}$$

which is only zero when $\mathbf{a} \wedge \mathbf{b} \wedge \mathbf{c} = 0$. (what does the Cauchy Binet theorem say about this determinant?) Use this fact to show $K = \frac{\mathbf{a}^2 \mathbf{b}^2 \mathbf{c}^2}{\mathcal{A}(\mathbf{a}, \mathbf{b}, \mathbf{c})}$. Conclude that a necessary condition for a quadrilateral to be cyclic is for opposite spreads to be equal. Explain why this is not a sufficient condition.

At this point we are finally able to state the relationship between the spreads of a triangle which does not involve the quadrances of the sides. It is called the *triple spread law*.

Theorem 4 Given a non-null, non-colinear triangle ABC with spreads s_A, s_B, s_C at the vertices of A, B, C respectively, then we have

$$(s_A + s_B + s_C)^2 = 2(s_A^2 + s_B^2 + s_C^2) + 4s_A s_B s_C.$$

In particular if $s_C = 1$ then $s_A + s_B = 1$; spreads are only additive in a right triangle.

Proof Using the spread law we know each quadrance $Q_A = (B - C)^2$, $Q_B = (C - A)^2$, and $Q_C = (A - B)^2$ of the triangle is proportional to the opposite spread by the same scaling factor, so that $\lambda s_A = Q_A$, $\lambda s_B = Q_B$, and $\lambda s_C = Q_C$. Plugging this into the cross law we can say

$$\lambda^2 4s_A s_B (1 - s_C) = \lambda^2 (s_A + s_B - s_C)^2.$$

After cancelling λ^2 from both sides and rearranging we achieve the result.

A set of three numbers which satisfy the triple spread law is called a **spread triple**.

Exercise 6 This relationship we found is a cubic polynomial overall, but is quadratic in any variable if we fix the other two. Show that we can rewrite the spread law in asymmetric form as

$$s_C^2 - 2(s_A + s_B - 2s_A s_B)s_C + (s_A - s_B)^2.$$

Show that a triangle cannot have two right angles. Show that if all the quadrances of a triangle are equal then all of the spreads are $3/4$, conclude that we cannot find equilateral triangles over the real numbers in green or red geometry. If $c_A = 1 - s_A$, $c_B = 1 - s_B$, and $c_C = 1 - s_C$ then show we have $(c_A + c_B + c_C - 1)^2 = 4c_A c_B c_C$.

Before we get to the flashy application we will want wrap up the theory by introducing the spread polynomials $S_n(s)$. The triple spread law tells us (after some work) that if two spreads in a triangle are equal to s , then the third spread is 0 or $4s(1 - s)$. This motivates us to define $S_0(s) = 0$, $S_1(s) = s$, $S_2(s) = 4s(1 - s)$, and for $n \geq 2$ we recursively define S_n by insisting that $\{s, S_{n-1}(s), S_n(s)\}$ is also a spread triple. This means that if we have a sequence of vectors $\mathbf{v}_1, \dots, \mathbf{v}_n$ with $s(\mathbf{v}_0, \mathbf{v}_1) = s$ such that $\mathbf{v}_i = \mathbf{v}_{i-1} \mathbf{v}_{i-2} \mathbf{v}_{i-1}^{-1}$ for $i = 2, \dots, n$, then $s(\mathbf{v}_0, \mathbf{v}_n) = S_n(s)$. From this we get that the spread polynomials play nice with function composition, that is

$$S_n \circ S_m = S_{nm} \quad S_n \circ S_m = S_m \circ S_n \quad (S_n \circ S_m) \circ S_l = S_n \circ (S_m \circ S_l).$$

There are some extremely interesting number theoretic properties of these polynomials, and we do not have the time or space to detail them except in the form of an exercise.

Exercise 7 The result of this exercise was first demonstrated by M. Hirschorn at the University of New South Wales. Show that the spread polynomials satisfy the second order recurrence relation $S_n(s) = 2(1 - 2s)S_{n-1}(s) - S_{n-2}(s) + 2s$ for all $n \geq 2$. Use this to show that the spread polynomials have the explicit formula

$$S_n(s) = \frac{2 - (1 - 2s + \sqrt{-4s(1-s)})^n - (1 - 2s - \sqrt{-4s(1-s)})^n}{4}$$

for all $n \geq 0$ and that they have the generating function

$$\sum_{n=1}^{\infty} S_n(s)t^n = \frac{st(1+t)}{(1-t)^3 + 4st(1-t)}.$$

Redo all of this analysis for cross.

Now for the best part. We say that two triangles are **similar** if they have the same spreads. A quick check will confirm that triangle similarity is an equivalence relation, and so similarity partitions all triangles in the plane. We can use the triple spread law to count the number of equivalence classes of similar triangles, which is finite whenever the field we are working over is. This is equivalent to counting the number of rational points on the surface $(s_A + s_B + s_C)^2 - 2(s_A^2 + s_B^2 + s_C^2) - 4s_A s_B s_C = 0$, and then identifying any points whose coordinates are permutations of each other.

Here is some code which will prove that there are exactly 26,800 equivalence classes of similar triangles over the field with 401 elements. I did my best to optimize this, it takes about 5 seconds to compute for a small prime like 401, but clearly the time complexity of this algorithm is a mess.

```
for p in {401}:
    count = 0
    for s in range(1, p):
        for t in range(s, p):
            limit = 0
            #this surface is quadratic in any variable, sometimes it can terminate early
            for u in range(t, p):
                # only points with weakly increasing coefficients are checked so we can
                # avoid the redunancy of identifying points which are permutations of
                # eachother.
                if limit <= 1:
                    check = (s+t+u) * (s+t+u) - 2 * (s*s+t*t+u*u) - 4*s*t*u
                    if check % p == 0:
                        count += 1
                        limit += 1
            print("For " + str(p) + ", the number of types of similar triangles is: " +
                  str(count))
```

Notice that this technique is agnostic of the quadratic form we use to measure spreads, and it does not care about the dimension of the space we are working in: adding more dimensions can not create new types of similar triangles. A caveat is that this technique is technically counting all *possible* similar triangles, until we have shown that every triple that satisfies the triple spread law is actually realized as the spreads of some triangle, which we have not done. It is an open question (to the author at least) what a generating function of these values for each finite field would be. As far as I can tell, the associated sequence of values for fields of prime size does not appear on the OEIS.

A simple moment of reflection about the existence of scalene and isosceles triangles should be able to convince the reader that if we are randomly selecting triangles in a plane, the resulting probability distribution induced on the equivalence classes is not uniform. Two more open questions are to determine the distribution of probabilities on these equivalence classes, and what is the probability that two triangles are similar if you are sampling with or without replacement.